Burugapalli Rohan

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AI & Backend Engineer

Education

Birla Institute of Technology and Science, Pilani — Dubai, UAE
B.Tech in Computer Science and Engineering — *Sep 2022 – Jul 2026*

- **Thesis:** Zero Trust Defences Against Social Engineering in Online Social Networks

Technical Skills

Languages: Python, TypeScript, SQL, Java, C++
Backend: FastAPI, Express, Prisma, Postgres, Redis, Celery, BullMQ, REST/OpenAPI, OAuth/JWT, nginx
AI/ML: PyTorch, Hugging Face Transformers, LangGraph, LlamaIndex, RAG, Chroma, Langfuse, LLM evaluation
Tools: Docker, GitHub Actions, pytest, Mocha, Vitest, Git

Experience

AI/ML Engineer Intern — Remote
Tolour — *Jan 2026 – Jul 2026*

- Built the FastAPI and Redis backend for an LLM assistant that reads and edits records via natural language
- Split the assistant into a planner that decides what data a request needs and a separate executor that accesses and acts on it, so one wrong decision cannot pass through unchecked
- Designed 5 layers of access control (role permissions, PII field blocking, record-level ownership checks, confirm-before-write, hard block on unauthorised data), all enforced in code rather than left to the model
- Wrote 8 isolated test suites covering every operation across two user roles, catching permission leaks, fabricated answers and wrong refusals before production
- Added logging, Langfuse tracing and Redis rate limiting; contributed to a LangGraph + LiveKit voice agent

Design Intern — Dubai, UAE
Everest DG & EIT Global — *Jun 2024 – Aug 2024*

- Built interactive web and product prototypes for team and client review

Embedded Systems Intern — Hyderabad, India
Vector Techniques — *Jun 2023 – Aug 2023*

- Collected BLDC motor speed data under varying loads using embedded C and sensors, then analysed it to identify optimal operating parameters and reduce performance variance

Projects

PR Review Agent | *Python, FastAPI, LangGraph, Postgres, Redis, Next.js* | github

- LangGraph agent that reviews GitHub pull requests and returns findings pinned to file and line ranges; GitHub access is read-only at the transport layer, enforced before a socket opens
- GitHub OAuth, permission-based RBAC, per-user sliding-window rate limits in Redis, and audit rows committed in the same transaction as each action; every security claim backed by a test
- Evaluated on 215 cases against a baseline: found 76% of real reverted bugs and localised all of them, stayed silent on 71% of clean PRs at $0.10 per review; contained 20/20 live prompt-injection attacks

Beta – Reminder-First Habit Tracker | *TypeScript, Express, Prisma, Postgres, Redis, BullMQ* | github

- Full-stack habit tracker that pushes a notification when each habit is due, even with the app closed; React PWA and Express API share a TypeScript core so due times match in each user's timezone
- Reminder engine: each habit change triggers a BullMQ job that replans the user's reminders, and a worker claims due ones with Postgres `SKIP LOCKED` for at-most-once delivery, re-checking completions and snoozes before sending
- Short-lived access tokens with rotating refresh cookies, where reusing an old token signs out the whole session; Redis caching and per-user rate limits; CI tests against real Postgres and Redis
- Cloud exports to S3 via presigned URLs and password-reset email via SES (AWS SDK v3), developed and tested in CI against a local AWS emulator (Floci + Mailpit); Docker Compose dev stack

EvalHarness – LLM Tool-Use Evaluation | *Python, FastAPI, Celery, Postgres, Next.js* | github

- Config-driven harness grading how LLMs call tools in multi-turn workflows with deterministic assertions; each scenario runs k times for pass^k consistency, with results tied to a config hash
- Across 4 models and 100 attempts, caught a model obeying a prompt injection that a regex check scored clean, and showed no pair of models was separable at n=25